

Matplotlib Cheat Sheet with Examples and Outputs

1. Basic Plotting

Function	Description	Example	Output
plt.plot()	Line plot	plt.plot([1,2,3,4])	Line graph
plt.scatter()	Scatter plot	plt.scatter([1,2,3],[4,5,6])	Dots at points
plt.bar()	Bar chart	plt.bar(['A','B','C'], [3,7,5])	Bars for A,B,C
plt.hist()	Histogram	plt.hist(np.random.randn(1000))	Default histogram
plt.pie()	Pie chart	plt.pie([10,20,30])	Pie with 3 slices

2. Customization

Function	Description	Example	Output
plt.title()	Add title	plt.title('My Graph')	Title displayed
plt.xlabel()	X-axis label	plt.xlabel('Time (s)')	Label below x-axis
plt.ylabel()	Y-axis label	plt.ylabel('Value')	Label left of y-axis
plt.legend()	Add legend	plt.legend(['Data 1','Data 2'])	Legend box
plt.grid()	Add grid	plt.grid(True)	Grid lines

3. Styling

Function	Description	Example	Output
plt.style.use()	Set style	plt.style.use('ggplot')	ggplot-style plot
color=	Set color	plt.plot(y, color='red')	Red line
linestyle=	Line style	plt.plot(y, linestyle='--')	Dashed line
marker=	Marker style	plt.plot(x, y, marker='o')	Dots at points

4. Multiple Plots

Function	Description	Example	Output
plt.subplot()	Multiple plots	plt.subplot(2,1,1)	Top plot of 2 rows
plt.subplots()	Subplot grid	fig, ax = plt.subplots(2,2)	2x2 grid
plt.tight_layout()	Adjust spacing	plt.tight_layout()	No overlap

5. Advanced Plots

Function	Description	Example	Output
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plt.fill_between()	Fill between lines	plt.fill_between(x, y1, y2)	Shaded region
plt.contour()	Contour lines	plt.contour(X,Y,Z)	Contour map
plt.imshow()	Display image	plt.imshow(img)	Image display

6. 3D Plotting

Function	Description	Example	Output
ax.plot3D()	3D line	ax.plot3D(x, y, z)	3D line plot
ax.scatter3D()	3D scatter	ax.scatter3D(x, y, z)	3D dots
ax.plot_surface()	3D surface	ax.plot_surface(X, Y, Z)	Surface plot

7. Annotations & Text

Function	Description	Example	Output
plt.text()	Add text	plt.text(1,1,'Point A')	Text label
plt.annotate()	Add arrow	plt.annotate('Max', xy=(2,4))	Arrow annotation
plt.axhline()	Horizontal line	plt.axhline(y=0)	Line across plot

8. Saving & Display

Function	Description	Example	Output
plt.savefig()	Save file	plt.savefig('plot.png')	Image saved
plt.show()	Display	plt.show()	Opens window
plt.close()	Close	plt.close()	Closes figure